

- ① By KVL, the total voltage across series is sum of voltage drops, while current is constant.

$$V_{\text{tot}} = \sum V_i$$

$$I = I_1 = I_2 = \dots = I_n$$

Note that $V = IR$

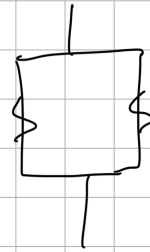
$$\text{so } I_{\text{tot}} R_{\text{tot}} = \sum I_i R_i$$

$$I R_{\text{tot}} = \sum I R_i = I \sum R_i$$

$$R_{\text{tot}} = \sum R_i$$

This generalizes to n resistors using induction.

- ② Note that parallel components have the same voltage



$$V = V_1 = V_2$$

$$I R_1 = I R_2$$

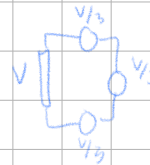
$$\text{so } I_1 = \frac{V}{R_1} \quad I_2 = \frac{V}{R_2}$$

while current splits

$$\text{By KCL: } I = I_1 + I_2$$

$$\frac{V}{R_{\text{eq}}} = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$



Solution:

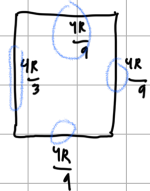
Start from $n/3$:



split this R into



split this $2R$ into



Our smallest resistor is $\frac{4R}{4}$.

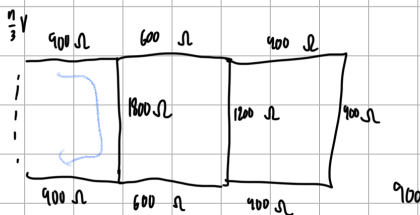
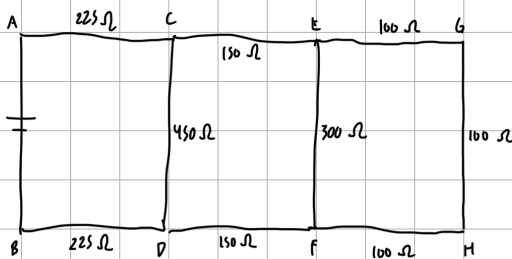
$$\text{so } \frac{4R}{4} = 100 \Rightarrow R = \frac{900}{4} = 225$$

$$\frac{2R}{3} = \frac{225(2)}{3} = 150$$

Hence:

$\frac{n}{3}$	AC, BD
$\frac{n}{4}$	CE, DF
$\frac{n}{27}$	EG, FH, FI

③



900Ω

$$6 \rightarrow 2 \rightarrow$$

$$2.25$$

$$1.5$$

$$1$$

$$9 \rightarrow 4.5 \rightarrow 2.25$$

$$4.5$$

$$3$$

$$1$$

$$7.5 \rightarrow 3.7$$

$$2.25$$

$$1.5$$

$$1$$

$$6$$